

THE VAST NETWORK

Key Concepts of Computer Networking

Computer Networking links computers for shared resources through the internet. Networking requires computers to access multiple networks and share their resources for computer operations, including web browsing, sending and retrieving emails, sharing files and images, and downloading music.

Organizations such as companies, schools, and government agencies enforce the use of single-user or general-purpose computer networks for word processing, scientific computation, and control processing.

Network Categories based on Geography

Networks are categorized based on their geographic scope:

- **Personal Area Network (PAN):** The smallest type of network with a limited span interconnects compatible personal devices. Examples include wireless earphones-to-smartphone and computer-to-printer connections.
- **Local Area Network (LAN):** This is used to interconnect and share computer resources in an establishment such as a corporate building, house, or several buildings in a limited area.
- **Campus Area Network (CAN):** An enterprise network covering multiple buildings in a campus environment, including universities and large organizations.
- **Metropolitan Area Network (MAN):** This covers multiple city buildings, traffic lights, and parking meters, connecting wirelessly as a metropolitan area network.
- **Wide Area Network (WAN):** A much larger network covering vast areas, such as cities, provinces, and countries. WAN can also be a bigger network consisting of LANs and MANs.

Network Architecture

It is a diagram displaying the structure of network devices and services to serve the clients' connectivity needs.



Source: <https://www.cisco.com/c/en/us/solutions/enterprise-networks/what-is-network-architecture.html>

Figure 1. A sample network architecture

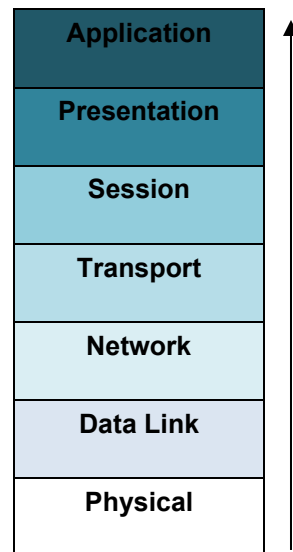
Common Types of Network Architecture

Fulfilling the needs of the client is the biggest goal of computer networks. Here are the three common types of enterprise networks:

1. **Access networks** in campuses and branches bring users and devices together, such as linking employees in a corporate building.
2. **Networks for data centers** in a company link multiple servers with data and applications and make them easily accessible to users.
3. **Wide-area networks (WANs)** connect users to programs and services, such as hospital employees connecting to health applications.

The Open Systems Interconnection Model

The Open Systems Interconnection (OSI) reference model of the International Organization for Standardization was developed in 1984 as an open standard for every communication system to allow the interlinking of different types of networks.



Source: Beasley J., Nilkaew P. (2022). *Networking essentials: Sixth edition*. Pearson.

Figure 1. The seven layers of the OSI model

The OSI reference model has seven layers describing networking functions, from the physical network interface to the software application interface. OSI model aims to provide a structure for networking that secures compatibility in the network hardware and software and expedites the advancement of upcoming networking technologies.

The Layers of the OSI Model

1. **Physical layer** – gives electrical and mechanical connections to the network.
2. **Data link layer** – handles error recovery, flow control, and sequencing of terminals that are either sending or receiving. It is the media access control (MAC) layer where MAC addressing is defined.
3. **Network layer** – acts as the network controller that receives outgoing messages and combines messages and segments into packets with a header detailing the routing information.
4. **Transport layer** – liable for end-to-end delivery between devices. It deals with message integrity between source and destination and segmenting and reassembling packets.
5. **Session layer** – provides the control functions required to start, manage, and end connections to satisfy user requests.
6. **Presentation layer** – accepts and structures the messages for an application by converting the message from one code to another. It also processes data compression and encryption.
7. **Application layer**: It interacts with application programs with communication components such as a web browser and email. It is responsible for recording a message, understanding the request, and identifying the information needed to support it.

Summary of the OSI Layers

Layer	Function
Physical	Handles signal and media
Data Link	Assigned for the flow of data
Network	Provides addressing and routing decisions
Transport	Manages end-to-end delivery to ensure error-free packets
Session	Establishes, manages, and terminates sessions
Presentation	Manages protocol conversion and data translation
Application	Provides support for applications

Network Types and Devices

These are devices required for communication and interaction between components on a computer network, either wired or wireless.

Wired Network uses cabling and connectors to initiate network connections.

Wireless Network uses radio signals to initiate network connections. It is the most common home network configuration today.

Network Type	Advantages	Disadvantages
Wired	Faster network data transfer speed	Specialized tools for cable connections
	Inexpensive setup	Labor-intensive
	Not susceptible to outside interference	
Wireless	User mobility	Security issues
	Simple installation	Slower data transfer

Wired Network Devices

1. **LAN Ethernet Cable** – connects devices within a local area network. It is limited by length and durability specifications resulting in a bad signal if the cable is too long or of poor quality.
2. **Hub** – interconnects networking devices but risks the broadcast of data it receives to all devices connected to it.
3. **Switch** – initiates a direct connection from the sender to the destination bypassing the data traffic to ensure data privacy, unlike a hub. It is best for interconnecting networking devices.
4. **Cable Modem** – uses a cable connection to create a broadband network connection from a home network to the internet service provider (ISP). This setup needs a *splitter* to divide the cable TV from the home network.
5. **Servers** – provide resources, data, services, and programs to other computers or clients over a network. It is tasked to handle complex operations such as databases, emails, and shared files.
6. **Firewall** – a security device that monitors and secures incoming and outgoing network traffic based on established security policies. It is essentially the barrier between a private internal network and the public Internet.

Wireless Network Devices

1. **Wireless Access Point** – connects wireless devices and provides a connection to wired LANs. The data transfer speed for access points relies on the wireless technology of the clients.
2. **Wireless Router** – consists of a router, switch, and wireless access point that uses radio frequency to link networking devices. It is the most common way to connect wireless LANs to the ISP's access device.

3. **Wi-Fi Modem** – combines the functionality of a modem and a router, thus making it a convenient source for Internet connectivity.

Various Network Topologies

Network Topologies

LAN is described by the protocol and the topology used for network access. The **networking protocol** is the rule for users to exchange information. At the same time, the **network topology** is the physical and logical arrangement of nodes such as a switch or hub and connections in a network.

The most common topologies for LANs are as follows:

- **Point-to-Point** – the simplest network topology where two devices, like a computer or router, are connected directly. With this, the communication only flows between the two devices.
- **Token Ring** – Its deterministic nature ensures access to message transmissions at regular or fixed time intervals for each connected station.

A disadvantage is that if an error changes the token pattern, it might stop circulating, causing data traffic to stop. It is important to note that Ethernet technology has recently replaced token ring topology.

- **Bus** – occurs when computers share the media for data transmission using a coaxial cable, preventing data transmission from being extremely bandwidth efficient.

For example, if a computer prints a file, the data communication is only between the computer and the printer. But in a bus system, each connected device can see the computer's data traffic to the printer. The other devices wait for pauses or halts before establishing their transmissions.

It is one of the reasons why bus topologies are rarely used in modern computer networks.

- **Star** – the most common networking topology in LANs today, where all the devices connect to a central switch or hub.

For example, the hub or switch supplies the network connection if a computer needs to send data to the laser printer. If a hub is used, the data is sent to the hub, which forwards it to the printer.

But a hub is a multiport repeater that broadcasts and displays the data it receives to the devices connected. It also applies to all networking devices interconnected in a star network.

Additionally, to lessen unnecessary data traffic and isolate sections of a network, using a switch at the center of a star network is much more ideal than a hub.

- **Mesh** – it is when the networking devices are directly connected where several paths exist between the end devices.

Mesh topology can be divided into two types based on paths; **fully meshed** and **partially meshed**. If a direct path exists from each end device to other end devices in the network, it is a fully meshed topology. However, it is a partially meshed topology if multiple paths are observed between the end devices.

References:

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